

JAMES RUSE AGRICULTURAL HIGH SCHOOL

Mathematics (2 unit)

Year 12 Term 1 Assessment 2001

TIME ALLOWED: 85 Minutes

INSTRUCTIONS:

- Start each question on a new page
- All questions are of equal value
- Each question is to be **handed in separately**
- Marks may not be awarded for poorly arranged work.
- Standard integral tables are at the end of the paper.

QUESTION 1: (12 marks) Start this question on a new page

a) Integrate with respect to x :

i) $\cos\left(\frac{\pi}{2} - x\right)$

ii) $\frac{1}{2(3x+1)^2}$

iii) $\frac{x}{1+3x^2}$

b) Evaluate:

i) $\int_1^2 \frac{3x^2 - 5x + 7}{x} dx$

ii) $\int_0^{\frac{1}{4}} \left(\frac{\pi}{2} + e^{-4x}\right) dx$

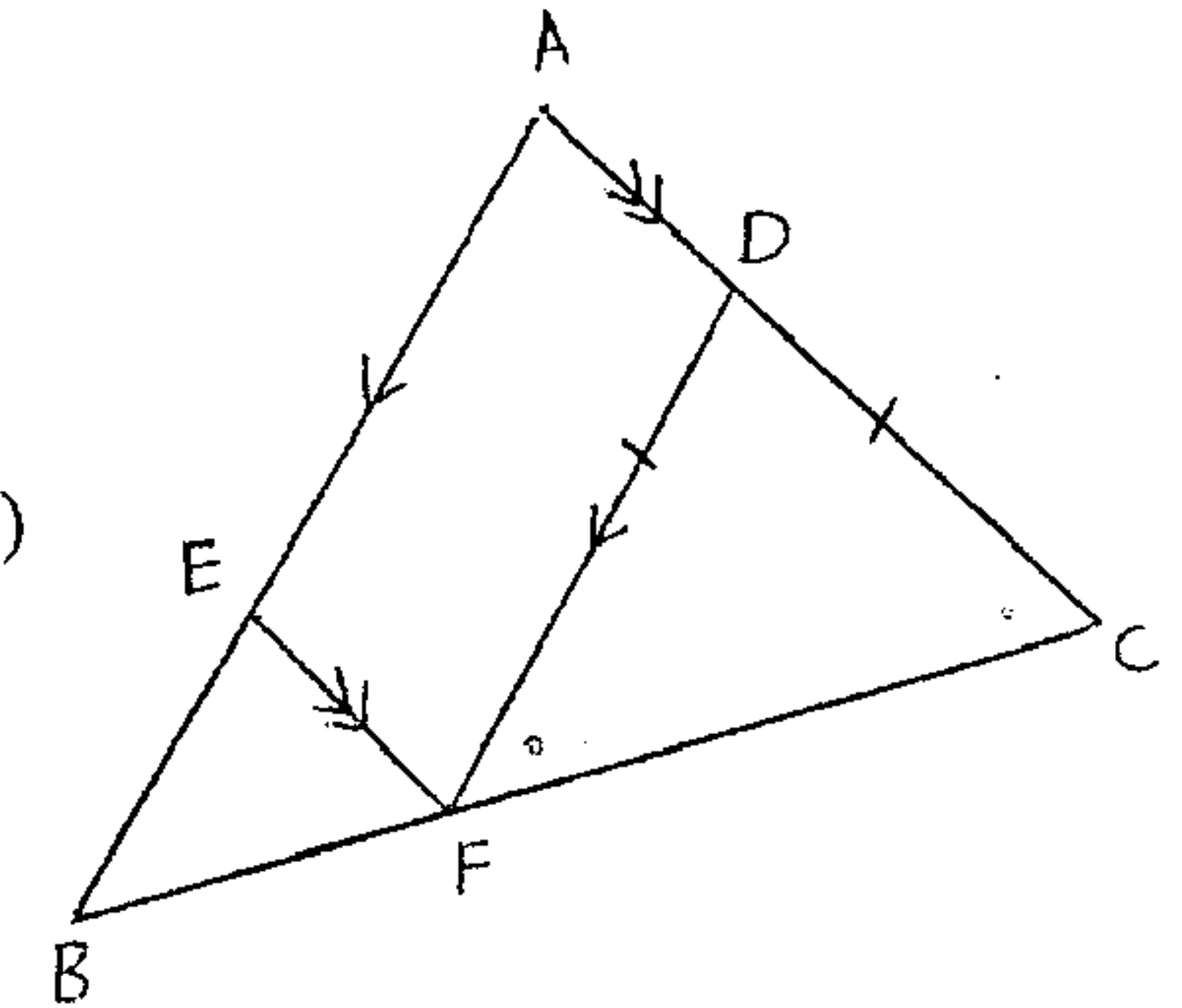
c) John receives \$5 pocket money each week when he is 5 years old, \$6 when he is 6, \$7 when he is 7 and so on. If he saves up all his pocket money for a car, how much will he have on his 17th birthday? (Do not include interest)

QUESTION 2: (12 marks) Start this question on a new page

a) Using Simpson's Rule with 5 function values to give an estimate of $\int_0^2 3^x dx$.
(Give answer correct to 3 significant figures)

b) In $\triangle ABC$, $DF=DC$, $DF \parallel AB$, $EF \parallel AC$

- i) Prove $\triangle ABC$ is isosceles (giving reasons)
- ii) If $\angle BAC$ is 3 times the size of $\angle ACB$, find $\angle ACB$.



c) On separate graphs sketch the following curves, showing their intercepts with the coordinate axes, and any asymptotes if they exist:

i) $x^2 + y^2 - 6x + 8y = 0$

ii) $\alpha) y = \frac{4-x}{x+1}$

$\beta)$ Using the graph, or otherwise, find all solutions to $\frac{4-x}{x+1} \leq 0$

QUESTION 3: (12 marks) Start this question on a new page

a) Find the limiting sum of the infinite series $\frac{1}{10^2} + \frac{2}{10^4} + \frac{4}{10^6} + \dots$

b) F.B.Jones invests \$5000 at an interest rate of 12.5% p.a. for 4 years compounded six monthly.

- i) Find, to the nearest dollar. The interest earned.
- ii) Find, to the nearest 0.1%, the annual interest rate that would bring the same return in 3 years (ie. If \$5000 was invested for 3 years and compounded monthly).

c) i) On the same graph, sketch $y = (x+1)^2 + 3$ and $y = 6x$, showing clearly any points of intersection and the x/y intercepts.

iii) Find the exact area bounded by these two graphs and the line $x = -1$.

QUESTION 4: (12 marks) Start this question on a new page

a) Evaluate $\sum_{r=1}^n 2(3^r)$

b) i) Prove $\Delta PQT \parallel \Delta PSR$ (giving reasons)

ii) Find length RS

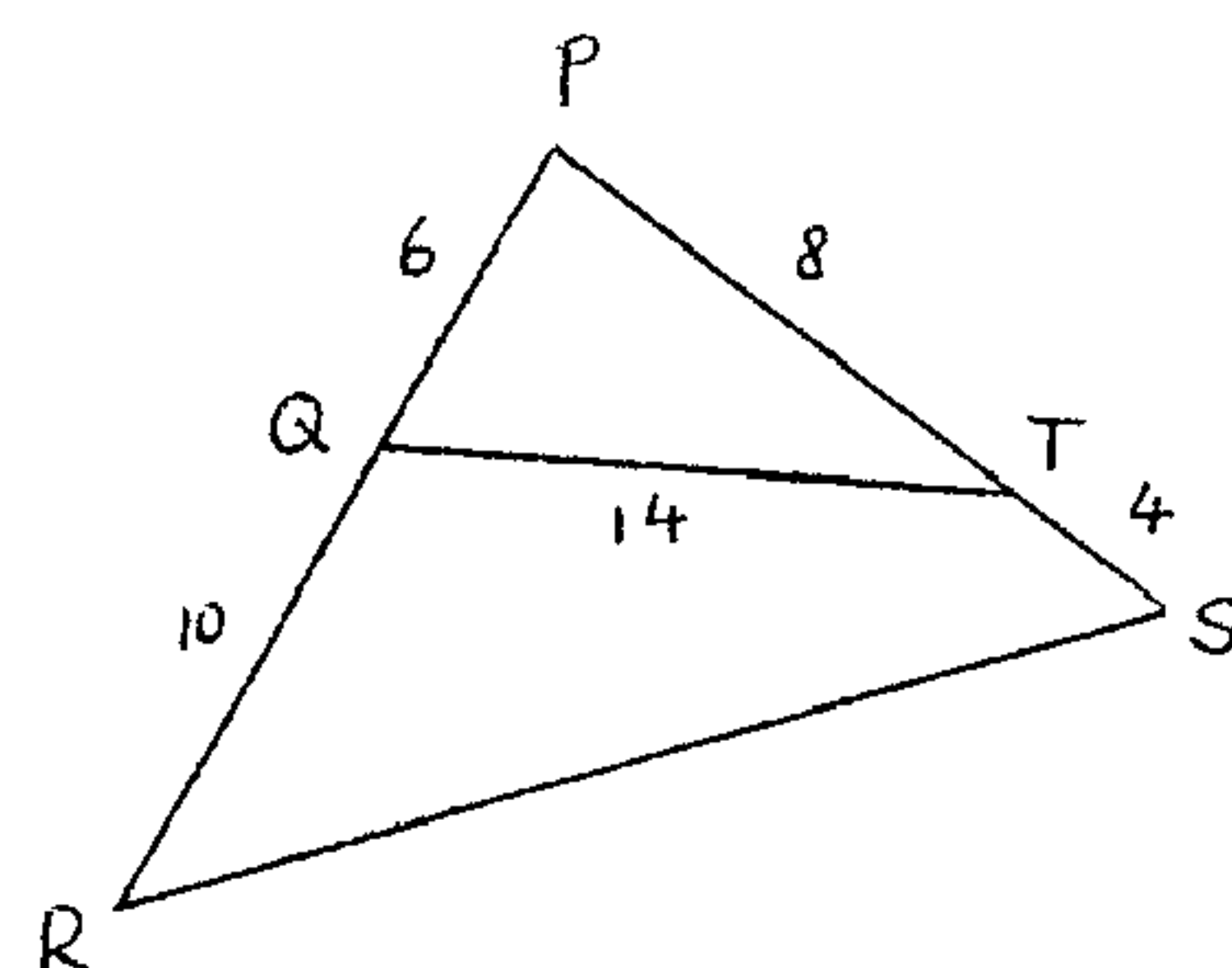
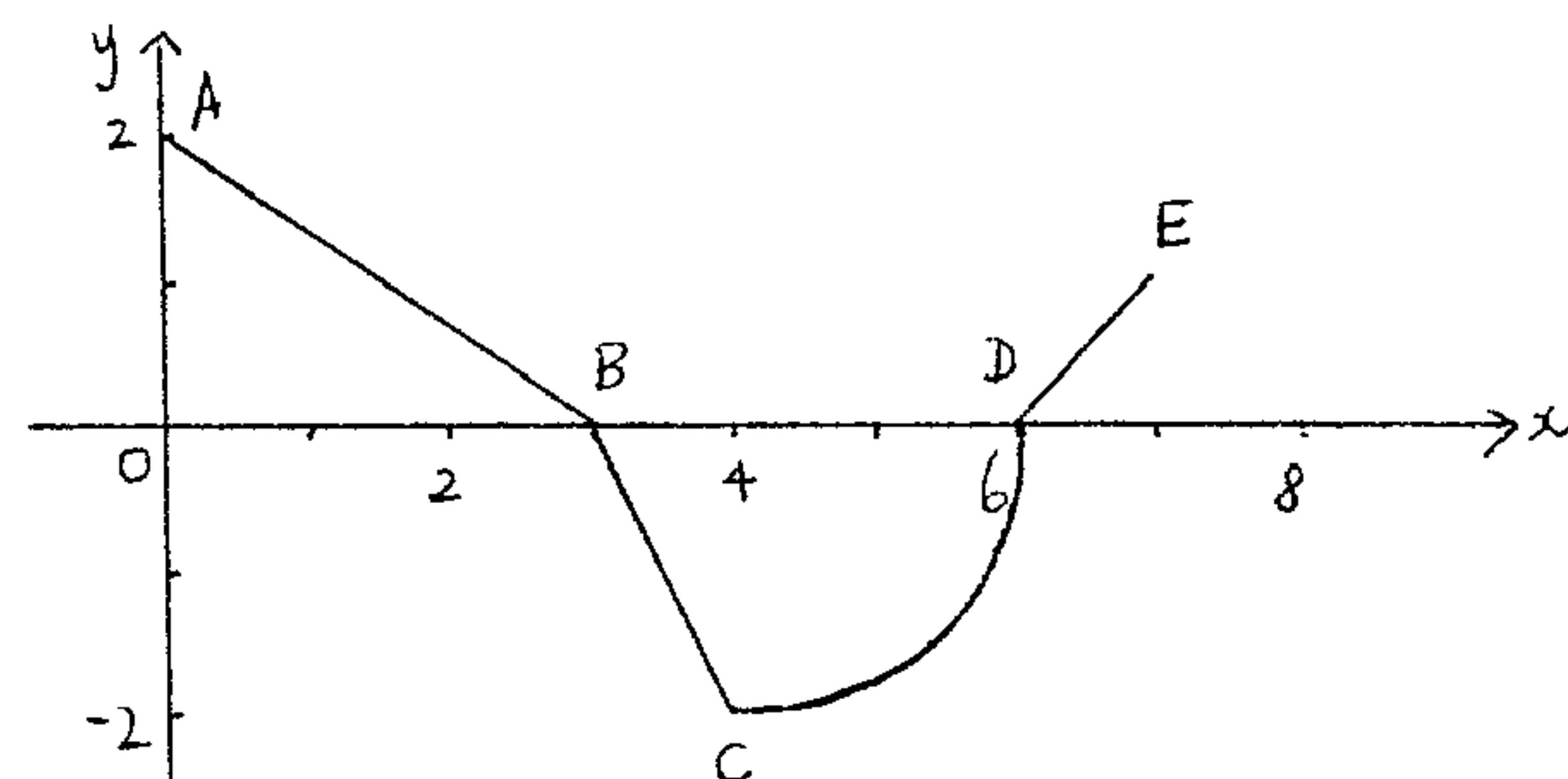


Diagram not to scale

c) The graph of the function $y = f(x)$ consists of the line segments AB, BC, DE and a quarter circle CD as shown below.

i) Evaluate $\int_0^7 f(x) dx$

ii) For what values of x in the domain $0 < x < 7$ is the function not differentiable.

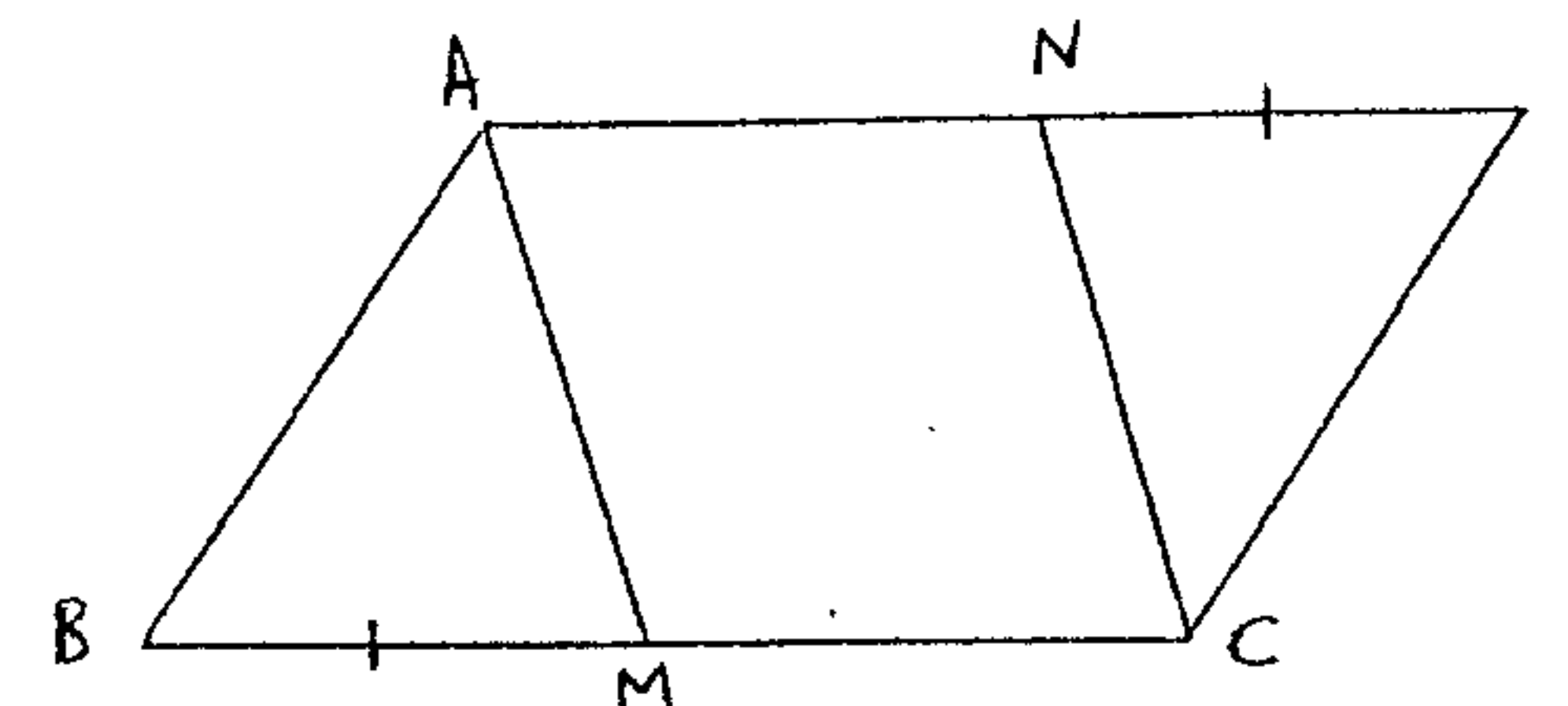


d) The cost of a new car is \$C. Its market value falls annually in geometric progression and is $\frac{C}{10}$ at the end of 10 years.

- Find the common ratio of the progression, correct to 4 decimal places.
- If the new car costs \$35 000, find the market value at the end of 5 years (to nearest dollar)

QUESTION 5: (12 marks) Start this question on a new page

a) ABCD is a parallelogram. $BM = ND$
Prove AMCN is a parallelogram

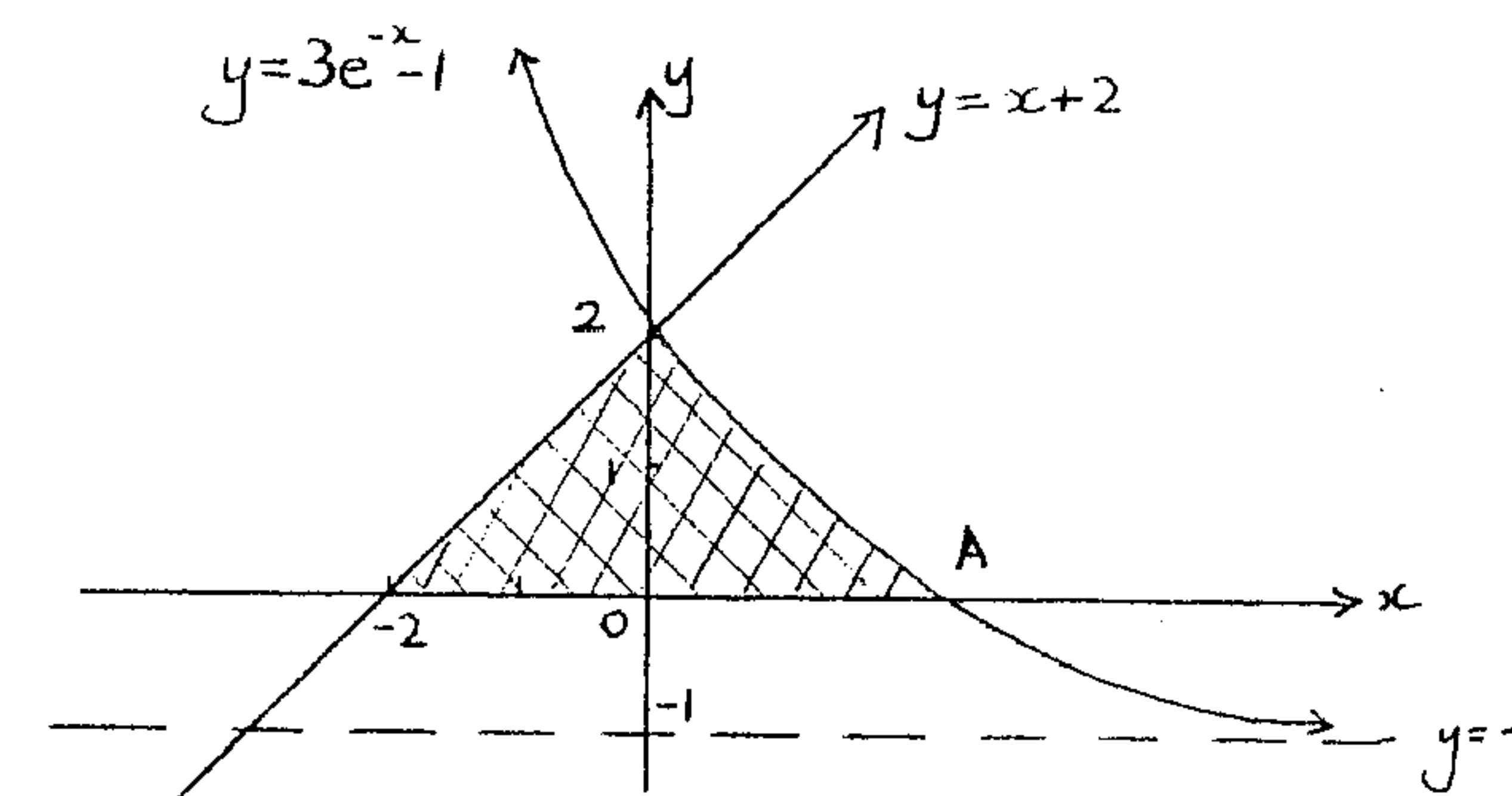


b) Sabrina bought a VCR for \$2000. She paid a deposit of \$150, and paid the balance off in equal monthly installments at 15% p.a. interest. If the loan was paid off after 24 months and Sabrina did not have to make any repayments for the first 3 months. Find,

- the amount owing after that 3 months,
- an expression for the amount owing after 4 months where M is the monthly repayment,
- derive an expression for the amount owing after n months,
- the amount of each monthly repayment (to nearest dollar).

c) The shaded area below shows the area bounded by $y = x + 2$, $y = 3e^{-x} - 1$ and the x -axis.

- Show that the point A has coordinate $(\ln 3, 0)$.
- Find the volume if the shaded area is rotated about the x -axis.



END OF PAPER

SOLUTIONS TO 2001 (20) MATHEMATICS

Question 1

a) (i) $\int \cos\left(\frac{\pi}{2} - x\right) dx$
 $= -\sin\left(\frac{\pi}{2} - x\right) + C$
 OR $= -\cos x + C$

b) (i) $\int_1^2 \frac{3x^2 - 5x + 7}{x} dx$
 $= \int_1^2 (3x - 5 + 7x^{-1}) dx$
 $= \left[\frac{3x^2}{2} - 5x + 7\ln x \right]_1^2$

(ii) $\int_0^{\frac{1}{4}} \left(\frac{\pi}{2} + e^{-4x} \right) dx$
 $= \left[\frac{\pi}{2}x - \frac{1}{4}e^{-4x} \right]_0^{\frac{1}{4}}$
 $= \left(\frac{\pi}{8} - \frac{1}{4e} \right) - \left(-\frac{1}{4} \right)$

(ii) $\int \frac{1}{2(3x+1)^2} dx$
 $= \frac{1}{2} \int (3x+1)^{-2} dx$
 $= \frac{1}{2} \cdot \frac{1}{3} (3x+1)^{-1} + C$
 $= -\frac{1}{6(3x+1)} + C$

$= (6 - 10 + 7\ln 2) - \left(\frac{3}{2} - 5 \right) = \frac{\pi}{8} - \frac{1}{4e} + \frac{1}{4}$
 $= 7\ln 2 - \frac{1}{2}$

OR $-\frac{1}{6(3x+1)} + C$

c) $5 \times 52 + 6 \times 52 + 7 \times 52 + \dots + 16 \times 52$
 $= 52(5 + 6 + 7 + 8 + \dots + 16)$
 $= 52 \left(\frac{12}{2} (5 + 16) \right)$
 $= \$ 6552$

(iii) $\int \frac{x}{1+3x^2} dx$
 $= \frac{1}{6} \ln(1+3x^2) + C$

NB: (2) each, total = 12 marks

Question 2

a)

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$f(x)$	3^0	$3^{\frac{1}{2}}$	3	$3^{\frac{3}{2}}$	3^2

$f(x) = 3^x$

$\therefore \int_0^2 3^x dx = \frac{1-0}{6} \left[3^0 + 4 \times 3^{\frac{1}{2}} + 3 \right] + \frac{2-1}{6} \left[3 + 4 \times 3^{\frac{3}{2}} + 3^2 \right]$
 $= \frac{1}{6} \left[4 + 12 + 4 \left(3^{\frac{1}{2}} + 3^{\frac{3}{2}} \right) \right]$
 $= 7.29 \text{ (3sf)} \quad (1/2)$

b) (i) Let $\angle ACF = x$

$\angle DFC = x$ (angles opp. equal sides are equal)

$\angle ADF = 2x$ (ext. angle of $\triangle DFC$)

$\angle EAD = 180 - 2x$ (co-interior angles supplementary, $AE \parallel DF$)

$\angle ABC = 180 - (180 - 2x + x)$ (angle sum of $\triangle ABC$)
 $= x$

$\therefore \triangle ABC$ is isosceles ($\angle ACB = \angle ABC = x$)

(1/3)

(ii) $x + x + 3x = 180$, $\therefore x = \frac{180}{5} = 36^\circ$

(1)

c) (i) $x^2 - 6x + y^2 + 8y = 0$

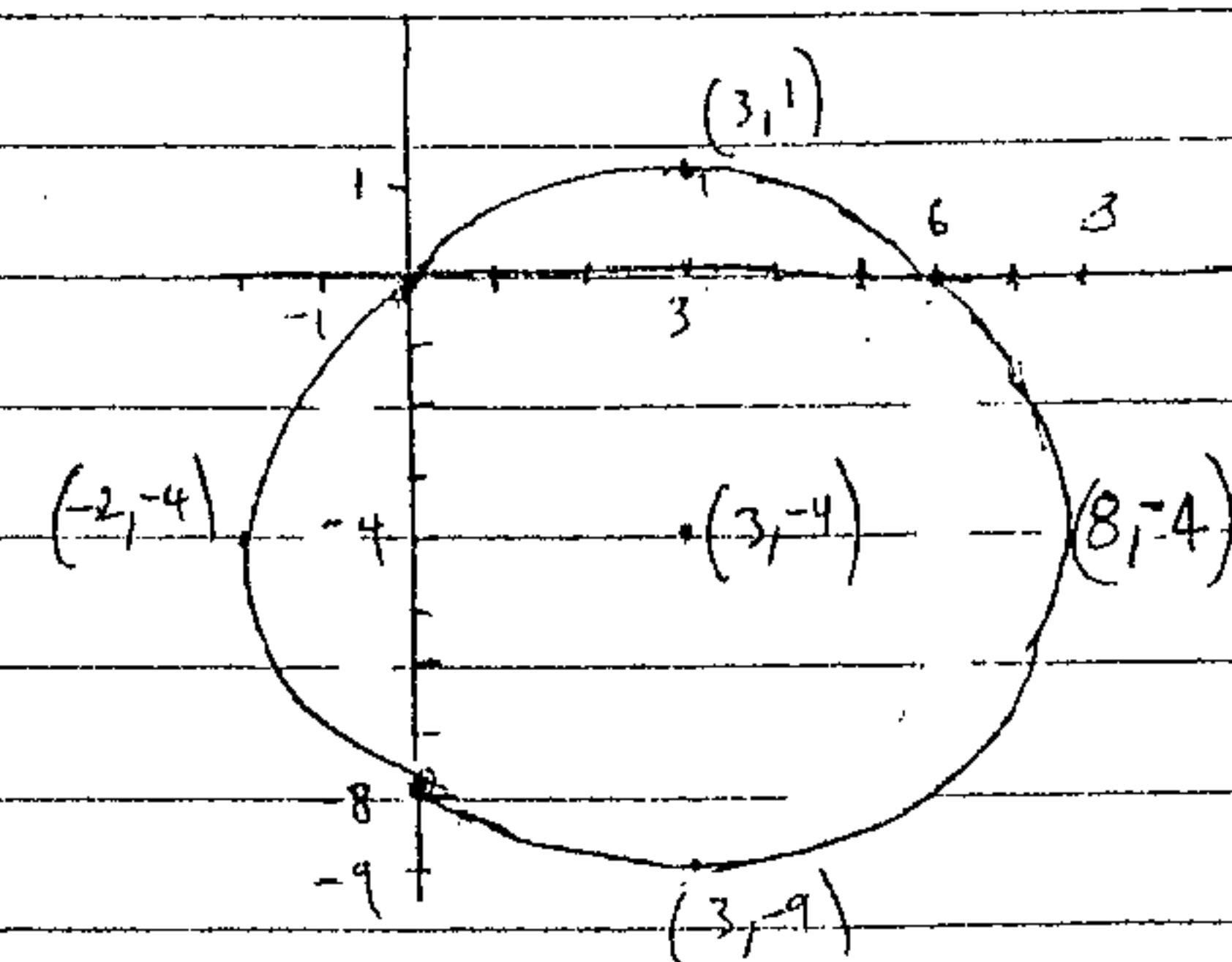
$x^2 - 6x + 9 + y^2 + 8y + 16 = 25$

$(x-3)^2 + (y+4)^2 = 25$

Centre = $(3, -4)$

Radius = 5

(2)

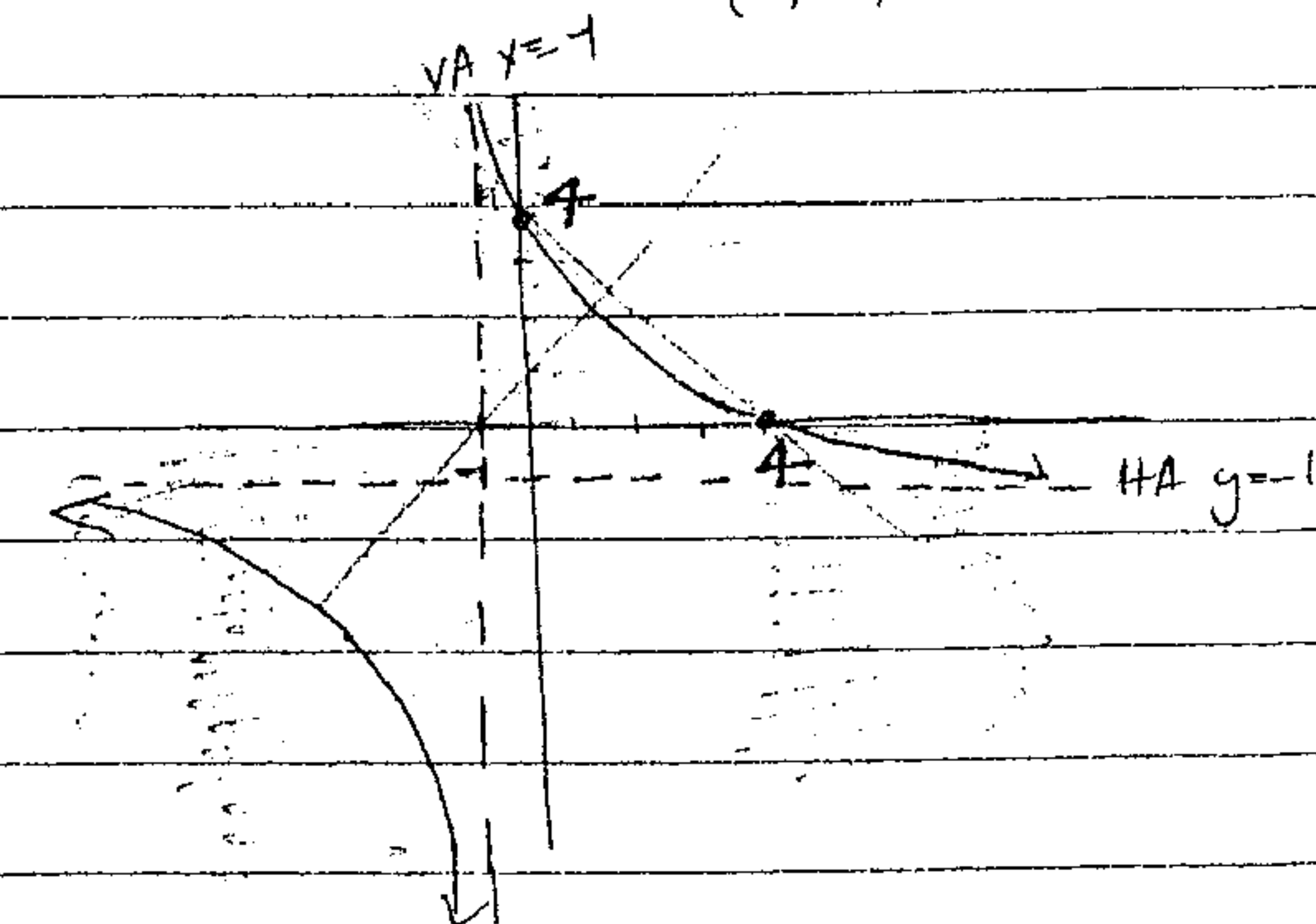


(ii) $y = \frac{4-x}{x+1}$

VA: $x = -1$
 HA: $y = -1$

y int ($x=0$): $y = 4$

x int ($y=0$): $x = 4$



$\begin{cases} x/y \text{ intercept } (\frac{1}{2} \text{ each}) = 1 \\ \text{VA \& HA } (\frac{1}{2} \text{ each}) = 1 \\ \text{Shape} = 1 \end{cases}$

(3)

$\beta) \frac{4-x}{x+1} \leq 0 \implies x < -1 \text{ or } x \geq 4$

1/2 each
(1)

Question 3

a) $a = \frac{1}{10^2}$, $r = \frac{2}{10^2}$, $\therefore S = \frac{\frac{1}{10^2}}{1 - \frac{2}{10^2}} = \frac{1}{98}$

(2)

b) (i) Interest = $5000(1.0625)^8 - 5000 = \$ 3121$

(1/2)

(ii) $(5000 + 3121) = 5000 \left(1 + \frac{R}{100} \right)^6$

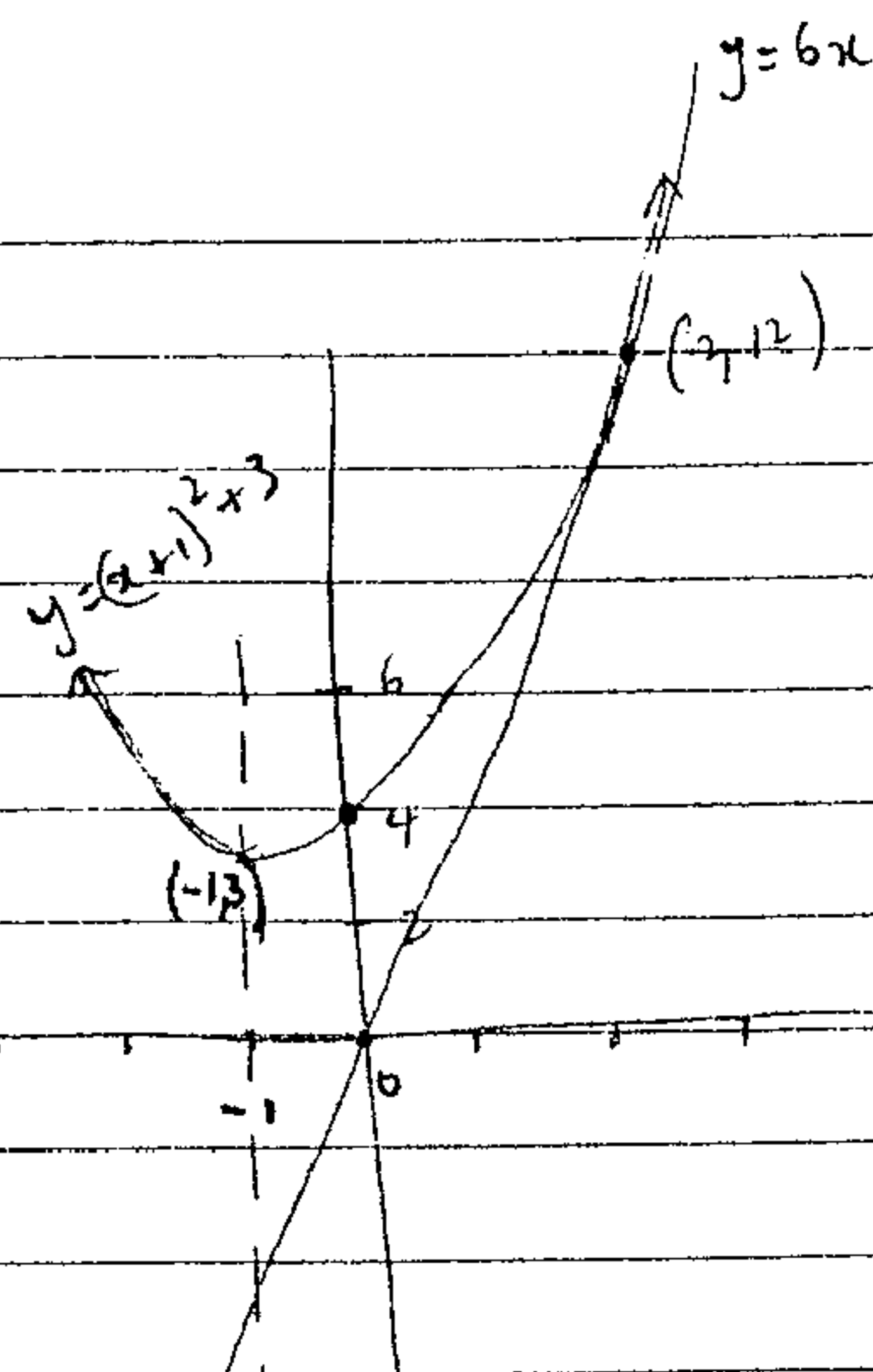
$\left(\sqrt[6]{\frac{8121}{5000}} - 1 \right) 100 = R$

$\therefore R = 8.4\%$ per six months

$\therefore R = 16.8\%$ per annum

(2)

c) (i)



$$y = (x+1)^2 + 3$$

$$y = 6x$$

$$\Rightarrow (2, 12) \text{ is pt of intersection}$$

(3) $\Rightarrow$ 1 for (2, 12)
 x & y intercepts ($\frac{1}{2}$ each) = 1.
 correct graphs ($\frac{1}{2}$ each) = 1

$$(ii) A = \int_{-1}^2 [(x+1)^2 + 3 - 6x] dx \quad 1$$

$$= \int_{-1}^2 (x^2 - 4x + 4) dx$$

$$= \left[\frac{x^3}{3} - 2x^2 + 4x \right]_{-1}^2$$

$$= \left(\frac{8}{3} - 8 + 8 \right) - \left(-\frac{1}{3} - 2 - 4 \right)$$

$$= \underline{9\frac{1}{3}}$$

Question 4

a) $T_n = 2(3^n)$ $a = 6$ $r = 3$

$$S_n = \frac{6(3^n - 1)}{3 - 1} = 3(3^n - 1)$$

b) (i) In ΔPQT & ΔPSR : $\frac{PT}{PR} = \frac{8}{16} = \frac{1}{2}$ } 1

$$\frac{PQ}{PS} = \frac{6}{12} = \frac{1}{2}$$

$\angle P$ is common

$\therefore \Delta PQT \sim \Delta PSR$ (2 adj sides in proportion & included angles equal)

(ii) $\frac{QT}{RS} = \frac{1}{2}$ (corresponding sides in same ratio)

ii $\frac{14}{RS} = \frac{1}{2} \Rightarrow RS = 28$

c) (ii) $\int_0^7 f(x) dx = \left(\frac{1}{2} \times 3 \times 2 \right) - \left(\frac{1}{2} \times 1 \times 2 \right) - \frac{1}{4} \pi \times 2^2 + \frac{1}{2} \times 1 \times 1$
 $= 3 - 1 - \pi + \frac{1}{2}$
 $= \frac{5}{2} - \pi$ 1 (1/2)

(ii) B, C, D

d) (i) Cost = C

Let V_n = value after n years.

$$V_1 = C \times \frac{1}{r}$$

$$V_2 = C \times \left(\frac{1}{r} \right)^2$$

$$\vdots$$

$$V_{10} = C \times \left(\frac{1}{r} \right)^{10}$$

But $V_{10} = \frac{C}{10} \Rightarrow \frac{C}{r^{10}} = \frac{C}{10}$ 1

$$r^{10} = 10$$

$$r = \sqrt[10]{10}$$

$$r = 1.2589 \text{ (4 dp.)}$$

(ii) If $C = 35000$

$$V_5 = 35000 \left(\frac{1}{1.2589} \right)^5 = \underline{\underline{\$11069}}$$

Question 5

a) $AN = AD - ND$

$$MC = CB - BM$$

Since $AD = CB$ (opp sides of parm $ABCD$ equal)

& $ND = BM$ (given)

Then: $AN = MC$

Also, $AN \parallel MC$ (opp. sides of parm $ABCD$ are parallel)

$\therefore ANCM$ is parm (one pair of opp. sides equal & parallel)

b) (i) After deposit, amount owing = $2000 - 150$
 $= \$1850$

Let A_n = amount owing after n months.

$$A_3 = 1850(1.0125)^3$$

$$= \underline{1920} \text{ (nearest dollar) OR } \$1920.24 \quad (1)$$

$$(ii) A_4 = 1850(1.0125)^4 - M$$

$$\text{OR } 1920.24 \times (1.0125) - M = 1944.25 - M \quad (1)$$

$$(iii) A_5 = (1850(1.0125)^4 - M)1.0125 - M$$

$$= 1850(1.0125)^5 - M(1 + 1.0125)$$

$$\therefore A_6 = 1850(1.0125)^6 - M(1 + 1.0125 + 1.0125^2) \quad (2)$$

$$A_n = 1850(1.0125)^n - M(1 + 1.0125 + 1.0125^2 + \dots + 1.0125^{n-4})$$

(iv) But $A_{24} = 0$

$$0 = 1850(1.0125)^{24} - M(1 + 1.0125 + 1.0125^2 + \dots + 1.0125^{20})$$

$$M = \frac{1850(1.0125)^{24}}{1 + 1.0125 + 1.0125^2 + \dots + 1.0125^{20}}$$

$$= \frac{1850(1.0125)^{24}}{\frac{(1.0125^{21} - 1)}{0.0125}} \quad (7)$$

$$= \underline{\underline{\$104.53}}$$

c) (i) $y = 3e^{-x} - 1$

$$y = 0$$

$$3e^{-x} - 1 = 0$$

$$e^{-x} = \frac{1}{3}$$

$$-x = \ln \frac{1}{3} \quad \therefore A = (\ln 3, 0) \quad (2)$$

$$x = -\ln \frac{1}{3}$$

$$x = \ln 3$$

(ii) $V = V_1 + V_2$

$$= \pi \int_{-2}^0 (x+2)^2 dx + \pi \int_0^{\ln 3} (3e^{-x} - 1)^2 dx$$

$$= \frac{1}{3} \pi \times 2^2 \times 2 + \pi \int_0^{\ln 3} (9e^{-2x} - 6e^{-x} + 1) dx$$

$$= \frac{8\pi}{3} + \pi \left[-\frac{9}{2}e^{-2x} + 6e^{-x} + x \right]_0^{\ln 3}$$

$$= \frac{8\pi}{3} + \pi \left[\left(-\frac{9}{2}e^{-2\ln 3} + 6e^{-\ln 3} + \ln 3 \right) - \left(-\frac{9}{2} + 6 \right) \right]$$

$$= \frac{8\pi}{3} + \pi \left(-\frac{1}{2} + 2 + \ln 3 + \frac{9}{2} - 6 \right)$$

$$= \frac{8\pi}{3} + \pi (\ln 3) \quad (3)$$